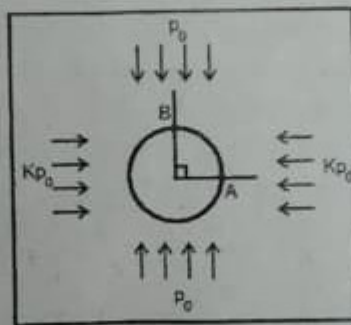


Instructions: 1) Answer any FIVE questions 2) Assume suitable data wherever required

Q. No	Question	Marks
1	Discuss briefly various data collection techniques for rock mass characterization in order to design of underground excavations in rock.	6
2	a) Define the modulus of deformation of rock mass. What are the techniques for determining in-situ stress in rock engineering projects? b) In a 1550-mm rock core run, the following rock pieces were recovered from a borehole: 53 mm, 108 mm, 125 mm, 75 mm, 148 mm, 320 mm, 68 mm, 145 mm, 35 mm, 19 mm and 134 mm. Find the Rock Quality Designation (RQD) and the core recovery ratio.	3 3
3	Describe briefly the design considerations in determining the appropriate location, size, shape, and alignment of underground tunnels and caverns in rock.	6
4	What are the various design methods for tunnels and caverns and state their applications. Explain the rock engineering design methodology for design of large underground excavations.	2+4
5	a) Give the classification of tunnels and large underground excavations based on their function and utilities. b) State the applications of Rock Mass Rating (RMR) and Q-system in underground rock tunnelling projects.	3 3
6	A tunnel is constructed in a biaxial far field stress (vertical) stress P_0 and horizontal stress KP_0 as shown in the figure. If the ratio of the tangential stress measured at the boundary points A and B is 4:1, Determine the value of K .	6



Note-1: Use the following formulas (Kirsch equations) wherever required

$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4\frac{a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta \right]$$